

# LSTM-Based Stock Price Trend Prediction

A Project Report on Time Series Forecasting

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## Abstract

This project implements a Long Short-Term Memory (LSTM) neural network model to forecast the next-day closing price trend of highly volatile stocks, using Tesla (TSLA) as the case study. The approach leverages deep learning techniques applied to time series data, enhanced by feature engineering that incorporates financial indicators. The primary goal was to demonstrate the full machine learning (ML) engineering pipeline, from data acquisition and preprocessing to model training, evaluation, and final deployment as an interactive web dashboard. The final model, incorporating the Close price, 20-Day Simple Moving Average (SMA), and Relative Strength Index (RSI), showed a noticeable improvement in predictive capability compared to a single-feature model, though the volatility of the target asset resulted in a high predicted price variance.

## 1 Introduction

Stock price forecasting remains one of the most challenging problems in financial time series analysis due to the non-linear, non-stationary nature of market data. Traditional statistical models often fail to capture the long-term dependencies inherent in stock movements. This project addresses this challenge by employing an LSTM network, a specialized type of Recurrent Neural Network (RNN) uniquely capable of learning sequence dependence and temporal patterns over long input sequences. The project focuses on predicting short-term trends (next-day closing price) and establishing a robust, end-to-end framework for financial predictive modeling.

## 2 Tools Used

The project utilized a powerful stack of Python libraries and frameworks:

- **Python 3.x:** Core programming language.
- **TensorFlow/Keras:** Deep Learning framework for building and training the LSTM network.
- **Pandas & NumPy:** Essential libraries for data manipulation and time series transformation.
- **yfinance:** API for reliable historical stock market data retrieval.
- **scikit-learn:** Used for data normalization (`MinMaxScaler`).
- **Matplotlib:** Visualization of trends and model predictions.
- **Streamlit:** Used for deploying the final model as an interactive web application (dashboard).

## 3 Steps Involved in Building the Project

### 3.1 Data Acquisition and Preprocessing

1. **Acquisition:** Historical stock data for TSLA was fetched using the `yfinance` API, spanning several years up to October 2025.
2. **Normalization:** Raw price data was scaled to  $[0, 1]$  using `MinMaxScaler`.
3. **Feature Engineering:** Technical indicators—20-Day Simple Moving Average (SMA) and 14-Day Relative Strength Index (RSI)—were calculated and added as input features.

4. **Sequence Creation:** The time series was transformed into the LSTM format: 180 previous trading days were used to predict the 181st day's price.
5. **Train/Test Split:** Data was split chronologically (80% for training, 20% for testing).

### 3.2 Model Development and Training

1. **Architecture:** A Sequential model was defined with two stacked LSTM layers (50 units each) and Dropout layers (0.2). Input shape was set to (180 timesteps, 3 features).
2. **Compilation:** Compiled using the Adam optimizer and Mean Squared Error (MSE) loss function.
3. **Training:** The model was trained over 25 epochs.

### 3.3 Evaluation and Deployment

1. **Evaluation:** Predictions were inversely scaled and evaluated using Root Mean Squared Error (RMSE).
2. **Serialization:** The trained model weights (`lstm_model_multi.keras`) and scaler object (`scaler_multi.pkl`) were saved.
3. **Deployment:** An interactive dashboard was created using Streamlit (`app.py`).

## 4 Conclusion

The LSTM model successfully learned the sequential dependencies in the TSLA stock data, demonstrating a functional deep learning forecasting pipeline. The addition of technical indicators (SMA and RSI) improved performance over the baseline single-feature model.

Despite being trained on recent data, the final prediction showed a significant discrepancy between the last recorded price (\$438.97) and the predicted price (\$339.56). This difference ( $\approx$  \$99) is attributed to the model interpreting the recent sharp price surge as unsustainable, leading to a strong prediction of **mean reversion** toward the longer-term average. While the model provides a robust technical framework, achieving higher prediction accuracy would require incorporating external factors like sentiment analysis or economic indicators. The project successfully met all goals, culminating in a fully deployed, interactive predictive dashboard.